This assignment will not be graded and is only for practice.

Level 1

1) Consider a function $T : V \to W$, where V and W are the vector spaces and following statements: ***1 point***

Statement 1: $T(v_1 + v_2) = T(v_1) + T(v_2)$, where $v_1, v_2 \in V$.
Statement 2: $T(cv_1) = cT(v_1)$, where $x \in V$ and $c \in \mathbb{R}$.

If T is a linear transformation then T satisfies

- statement 1 only.
- statement 2 only.
- both the statements 1 and 2.
- neither statement 1 nor statement 2.

No, the answer is incorrect.
Score: 0

Accepted Answers:

both the statements 1 and 2.

2) Which of the following is/are a linear transformation. ***1 point***

- $T : \mathbb{R}^2 \to \mathbb{R}^3, T(x, y) = (x, x + y, y + 1)$
- $T : \mathbb{R}^2 \to \mathbb{R}^2, T(x, y) = (xy, x + y)$
- $T : \mathbb{R}^2 \to \mathbb{R}^3, T(x, y) = (x + y, y, 2x + y + 1)$
- $T : \mathbb{R}^2 \to \mathbb{R}^3, T(x, y) = (x + y, y, 2x + y)$
- $T : \mathbb{R}^2 \to \mathbb{R}^2, T(x, y) = (\frac{x}{2}, \frac{y}{2})$
- $T : \mathbb{R}^2 \to \mathbb{R}^2, T(x, y) = (\frac{x}{2}, \frac{xy}{2})$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$T : \mathbb{R}^2 \rightarrow \mathbb{R}^3, T(x, y) = (x + y, y, 2x + y)$
$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2, T(x, y) = (\frac{x}{2}, \frac{y}{2})$

3) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ be a linear transformation. Then which of the following options is /are true? ***1 point***

- [ ] $T(v_1 + v_2) = T(v_1) + T(v_2)$, where $v_1, v_2 \in \mathbb{R}^3$.
- [ ] $T(0, 0, 0) = (0, 0, 0, 0)$
- [ ] $T(0, 0, 0) = (1, 1, 1, 1)$
- [ ] $T(3(1, 2, 0)) = 3T(1, 2, 0)$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$T(v_1 + v_2) = T(v_1) + T(v_2)$, where $v_1, v_2 \in \mathbb{R}^3$.
$T(0, 0, 0) = (0, 0, 0, 0)$
$T(3(1, 2, 0)) = 3T(1, 2, 0)$

4) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a linear transformation, defined as $T(x, y, z) = (x + y, y + z)$. Then which of the following options is /are true? ***1 point***

- [ ] $T(0, 0, 0) = (0, 0)$
- [ ] $T(1, 2, 3) = (3, 5)$
- [ ] $T(0, 1, 1) = (1, 2)$
- [ ] $T(1, 1, 1) = (3, 2)$
- [ ] T is injective.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$T(0, 0, 0) = (0, 0)$

$T(1,2,3) = (3,5)$
$T(0,1,1) = (1,2)$

5) Choose the correct set of options: **1 point**

- [ ] T is a one to one linear transformation if and only if there does not exist any $v = 0 \in V$ so that $T(v) = 0$.
- [ ] If $T : \mathbb{R}^2 \to \mathbb{R}^2$ is a surjective linear transformation, then it cannot be injective.
- [ ] If $T : \mathbb{R} \to \mathbb{R}$ is a linear transformation which is not injective, then T must be 0, i,e., $T(v) = 0$ for all $v \in \mathbb{R}$.
- [ ] If there exists some non-zero vector $v \in V$, such that $T(v) = 0$, for a linear transformation $T : V \to W$, then T cannot be an isomorphism.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is a one to one linear transformation if and only if there does not exist any $v = 0 \in V$ so that $T(v) = 0$.
If $T : \mathbb{R} \to \mathbb{R}$ is a linear transformation which is not injective, then T must be 0, i,e., $T(v) = 0$ for all $v \in \mathbb{R}$.
If there exists some non-zero vector $v \in V$, such that $T(v) = 0$, for a linear transformation $T : V \to W$, then T cannot be an isomorphism.

6) If $T : \mathbb{R}^2 \to \mathbb{R}^2$ is a linear transformation, such that $T(x, y) = (x, 0)$, then which of the following options is true? **1 point**

- ( ) T is both one to one and onto.
- ( ) T is one to one, but not onto.
- ( ) T is onto, but not one to one.
- ( ) T is neither one to one nor onto.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is neither one to one nor onto.

7) If $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation, such that $T(x, y) = (x + y, x - y)$, then which of the following options is true? ***1 point***

- T is both one to one and onto.
- T is one to one, but not onto.
- T is onto, but not one to one.
- T is neither one to one nor onto.

No, the answer is incorrect.
Score: 0

Accepted Answers:

T is both one to one and onto.

8) Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation, such that $T(2, 1) = (3, 2)$ and $T(1, 3) = (0, 0)$. Which of the following options is true? ***1 point***

- $T(x, y) = \frac{1}{5}(3x, 2y)$
- $T(x, y) = (x + 1, y + 1)$
- $T(x, y) = \frac{1}{5}(9x - 3y, 6x - 2y)$
- T cannot be determined from the given information.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$T(x, y) = \frac{1}{5}(9x - 3y, 6x - 2y)$

9) Consider the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined as $T(v) = Av$, where $v = \begin{bmatrix} x \\ y \end{bmatrix}$, and $A = \begin{bmatrix} 3 & 2 \\ 4 & 5 \end{bmatrix}$. Which of the following options is true? ***1 point***

- ○ T is an isomorphism.
- ○ T is one to one, but not onto.
- ○ T is onto, but not one to one.
- ○ T is neither one to one nor onto.

No, the answer is incorrect.
Score: 0

Accepted Answers:

T is an isomorphism.

Level 2

10) Let $S : V_1 \to V_2$ and $T : V_2 \to V_3$ be two linear transformations. Let us define $T \circ S : V_1 \to V_3$ by $T \circ S(v) = T(S(v))$. Choose the correct set of options. ***1 point***

- ☐ If $T \circ S$ is injective, then T must be injective.
- ☐ If $T \circ S$ is injective, then S must be injective.
- ☐ If $T \circ S$ is surjective, then T must be surjective.
- ☐ If $T \circ S$ is surjective, then S must be surjective.

No, the answer is incorrect.
Score: 0

Accepted Answers:

If $T \circ S$ is injective, then S must be injective.
If $T \circ S$ is surjective, then T must be surjective.

11) If $T : \mathbb{R}^2 \to \mathbb{R}^3$ is a linear transformation defined by $T(x, y) = (2x + 3y, 5x - y, x + 6y)$. Which of the following options is true? ***1 point***

- ○ T is both one to one and onto.

- T is one to one, but not onto.
- T is onto, but not one to one.
- T is neither one to one nor onto.

No, the answer is incorrect.
Score: 0

Accepted Answers:

T is one to one, but not onto.

12) If $T : \mathbb{R}^3 \to \mathbb{R}^3$ is a linear transformation defined as follows, $T : \mathbb{R}^3 \to \mathbb{R}^3$ **1 point**

$T(x, y, z) = (a_1x + b_1y + c_1z, a_2x + b_2y + c_2z, a_3x + b_3y + c_3z)$. This can be represented as $T(v) = Av$, where $A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$, and $v = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$. Which of the following options are true?

- T is injective if and only if the system of linear equations $Av = 0$ has a unique solution.
- T is injective if and only if $rank(A) = 3$.
- If the system of linear equations $Av = b$ for any $b \in \mathbb{R}^3$, has a solution, then T must be surjective.
- If T is not surjective, then there exists some $b \in \mathbb{R}^3$, such that the system of linear equations $Av = b$ has no solution.

No, the answer is incorrect.
Score: 0

Accepted Answers:

T is injective if and only if the system of linear equations $Av = 0$ has a unique solution.
T is injective if and only if $rank(A) = 3$.
If the system of linear equations $Av = b$ for any $b \in \mathbb{R}^3$, has a solution, then T must be surjective.
If T is not surjective, then there exists some $b \in \mathbb{R}^3$, such that the system of linear equations $Av = b$ has no solution.

Your score is: 0/12